Yash Joshi

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EDUCATION

PURDUE UNIVERSITY — West Lafayette, IN
B.S. in Mechanical Engineering (First Year Program) — Expected Graduation: May 2029

- **GPA:** 3.94/4.00
- **Relevant Coursework:** Linear Algebra, Ordinary Differential Equations, Geometric and Annotation Modeling, Basic Mechanics I, EE Fundamentals I, Vertically Integrated Projects (VIP)

NEUQUA VALLEY HIGH SCHOOL — Naperville, IL
High School Diploma, Magna Cum Laude — August 2021 – May 2025

EXPERIENCE

Purdue Lunabotics — August 2025 - Present
Drivetrain Sub-team Member

- Utilized Solidworks CAD to design gearboxes and essential frame elements optimized for performance and traversal efficiency
- Reduced assembly time of gearbox systems by 20% through clear documentation and thoughtful design of frequently replaced and adjusted components
- Developed CAM within Autodesk Fusion for high precision 3-axis CNC milling of aluminum gearbox parts and waterjet cutting of plexiglass skid plate parts, with consideration of 5+ kg weight reduction goal
- Maintained detailed documentation and trade studies to be presented in multiple design reviews in an effort to inform and learn from industry partners and club members

ASME (American Society of Mechanical Engineers) — August 2025 - Present
Industrial Relations Representative

- Communicated with industry partners including Blue Origin, Formlabs, and Oshcut in an effort to gather meaningful sponsorships and vital organization funding
- Worked with industry representatives and fellow executive board members to put together information sessions and recruiting events for 300+ organization members

Racing Team Member

- Utilized Autodesk Fusion and MotorXP to design an experimental YASA Axial Flux Motor with a 3000 RPM running speed
- Analyzed key aspects of the motor mounting assembly in Autodesk Fusion simulation to ensure the efficiency and structural stability of the motor when mounted and running
- Manufactured motor components and assembly devices using 3D printing (FDM and Resin) and Autodesk Fusion CAM based 3-axis CNC milling

STEAM Dream Organization — August 2022 - May 2025
Senior Event Coordinator

- Utilized Autodesk Fusion and FDM 3D printing to create specialized STEM learning kits with the goal of engaging 60+ students with hands-on activities
- Led fellow volunteers in raising $2500+ towards student exposure to STEM concepts in underprivileged communities

TECHNICAL PROJECTS

Obstacle Detecting Miniature Car

- Developed an autonomous miniature car using a 3D printed chassis and Ultrasonic sensors driven by an Arduino
- Created a C++ program within the Arduino IDE to detect and avoid obstacles with 90% accuracy in testing

Musical Transcription Program

- Wrote a Python program to gather audio input of music and properly display the notes being played in real time
- Utilized the NumPy, PyAudio, and MusPy libraries to effectively reach timing and accuracy goals

TECHNICAL SKILLS

CAx/Analysis Tools: Siemens NX, Solidworks, Autodesk Fusion, ANSYS, MotorXP, Aras Innovator
Technical Proficiency: GD&T, FDM 3D Printing, Arduino, Microsoft Office Suite
Languages: Python, C/C++, MATLAB, English, Spanish, Telugu